Week 6: Attention Mechanisms and Self-Attention

Redesigned Question Bank — Set 2 (Questions 11–20)

SIS 3204Y(5) Natural Language Processing

Assessment design note: Numerical questions use only arithmetic explicitly supported by the Week 6 lecture: simple dot products or weighted sums. Softmax values are not required to be calculated manually.

Question 11 — Attention at Different Output Steps [8 marks]

Question: Why is it useful for the attention distribution to change when a decoder generates successive output words?

Sample Answer

Different output words can depend on different parts of the input sequence. Therefore, the most relevant source information can change from one decoding step to another.

Attention allows a new set of relevance scores and weights to be produced for each output step. The resulting context vector can therefore emphasise different input positions as the decoder generates different words.

Mark Breakdown

Criterion	Marks
Different output words require different source information	2
New relevance scores/weights can be produced	2
Context vector changes accordingly	2
Clear benefit to decoding stated	2
Total	**8**

Question 12 — Reading Attention Weights [8 marks]

Question: For the sentence “The bank approved the loan”, suppose the attention weights for the token “bank” are:

The = 0.10, bank = 0.20, approved = 0.35, the = 0.10, loan = 0.25.

(a) Which token receives the highest attention? [2]
(b) Which token receives the second-highest attention? [2]
(c) Verify that the weights sum to 1. [2]
(d) What do these weights tell us about the contextual representation of “bank”? [2]

Sample Answer

(a) “approved” has the highest weight: 0.35.
(b) “loan” has the second-highest weight: 0.25.
(c) 0.10 + 0.20 + 0.35 + 0.10 + 0.25 = 1.00.
(d) The representation of “bank” combines information from the sentence, with “approved” and “loan” making relatively strong contributions in this illustrative example.

Mark Breakdown

Criterion	Marks
Highest weight identified	2
Second-highest identified	2
Correct sum	2
Correct contextual interpretation	2
Total	**8**

Question 13 — From Scores to Weights [8 marks]

Question: Why are raw attention scores normally passed through softmax before they are used to combine information?

Sample Answer

Raw attention scores are relevance values and are not necessarily directly suitable as weights. Softmax converts them into positive attention weights whose values sum to 1.

This makes the weights easier to use as relative contributions when forming the context vector or contextual representation: more relevant positions receive larger weights and less relevant positions receive smaller weights.

Mark Breakdown

Criterion	Marks
Raw scores described as relevance values	2
Softmax produces positive weights	2
Weights sum to 1	1
Weights represent relative contributions	2
Connection to context/output	1
Total	**8**

Question 14 — Q, K and V Misconception [9 marks]

Question: A student says: “Q, K and V are identical because they all come from the same input sequence.” Explain why this statement is incorrect.

Sample Answer

In self-attention, Q, K and V may all originate from the same input representations, but they are produced using different learned projection matrices.

Q = XW_Q, K = XW_K and V = XW_V.

They also have different functions: Queries are used to seek relevant information, Keys are compared with Queries to determine relevance, and Values contain the information that is combined to create the output representation.

Mark Breakdown

Criterion	Marks
Statement correctly rejected	1
Same input origin acknowledged	1

Different learned projections	2
Correct equations	2
Different functional roles explained	3
Total	**9**

Question 15 — Why Scale the Dot Product? [8 marks]

Question: Explain why scaled dot-product attention divides QK^T by √d_k before applying softmax.

Sample Answer

When the dimensionality of the Query and Key vectors is large, their dot products can become large in magnitude. Large score differences can make softmax overly peaked, with most of the weight concentrated on very few positions.

Dividing the scores by √d_k reduces their scale before softmax. This helps keep the softmax behaviour and gradients in a more manageable range during training.

Mark Breakdown

Criterion	Marks
Large-dimensional dot products can become large	2
Effect on softmax identified	2
Division by √d_k reduces scale	2
Training/gradient benefit explained	2
Total	**8**

Question 16 — Simple Dot-Product Relevance [8 marks]

Question: A query vector is q=[1,1]. Three Key vectors are:
k1=[1,0], k2=[1,1], k3=[0,1].

(a) Calculate q·k1, q·k2 and q·k3. [6]
(b) Based only on these dot-product scores, which Key is most relevant to the Query? [2]

You do NOT need to calculate softmax.

Sample Answer

(a)
q·k1 = (1×1)+(1×0) = 1
q·k2 = (1×1)+(1×1) = 2
q·k3 = (1×0)+(1×1) = 1

(b) k2 is the most relevant because it has the largest dot-product score, 2.

Mark Breakdown

Criterion	Marks
q·k1 = 1	2
q·k2 = 2	2
q·k3 = 1	2
k2 correctly identified and justified	2

Total	**8**

Question 17 — Self-Attention Step by Step [10 marks]

Question: Describe the self-attention process in the correct order, starting from input token representations and ending with contextualised output representations.

Sample Answer

First, the input tokens are represented numerically. Learned linear transformations create Q, K and V. Queries are compared with Keys to produce relevance scores. In scaled dot-product attention, the scores are divided by $\sqrt{d_k}$. Softmax converts them into attention weights. The weights are then used to combine the Value vectors. The resulting weighted combinations are contextualised representations of the tokens.

Mark Breakdown

Criterion	Marks
Input representations	1
Q/K/V creation	2
Query-Key comparison	2
Scaling	1
Softmax	1
Weighted combination of Values	2
Contextual output	1
Total	**10**

Question 18 — Context Vector Calculation [8 marks]

Question: The attention weights for three encoder states are [0.5, 0.3, 0.2]. The corresponding hidden states are h1=[2,0], h2=[0,1] and h3=[1,1].

Calculate the context vector. The attention weights are already provided.

Sample Answer

c = 0.5h1 + 0.3h2 + 0.2h3

= 0.5[2,0] + 0.3[0,1] + 0.2[1,1]

= [1,0] + [0,0.3] + [0.2,0.2]

= [1.2, 0.5]

Mark Breakdown

Criterion	Marks
Correct weighted-sum expression	2
Correct contribution of h1	1
Correct contribution of h2	1
Correct contribution of h3	1
Correct final context vector [1.2, 0.5]	3
Total	**8**

Question 19 — Single-Head and Multi-Head Attention [10 marks]

Question: Compare single-head attention and multi-head attention. Your answer should include how the outputs of multiple heads are combined.

Sample Answer

Single-head attention performs one learned attention operation at the layer. Multi-head attention performs several attention operations in parallel, with each head using its own learned projections of Q, K and V.

Each head produces an output representation. The outputs from the heads are concatenated and then transformed using an output projection. Multiple heads provide the model with several learned attention patterns or representation subspaces rather than only one.

Mark Breakdown

Criterion	Marks
Single-head attention described	2
Parallel multiple heads described	2
Different learned projections	2
Outputs concatenated	2
Output projection/benefit explained	2
Total	**10**

Question 20 — Complete Week 6 Concept Map [12 marks]

Question: Explain the following progression in your own words:

Attention → Self-Attention → Q/K/V → Scaled Dot-Product Attention → Multi-Head Attention → Transformers.

Your answer should show how each concept leads to the next.

Sample Answer

Attention allows a model to assign different importance to different input positions rather than relying only on one fixed representation. Self-attention applies this idea within the same sequence so that each token can build a context-aware representation.

Q, K and V provide the mechanism for deciding relevance and combining information. Scaled dot-product attention specifies how Queries and Keys are compared, scaled and converted into weights that combine Values. Multi-head attention repeats this process in parallel using several learned projections.

These attention mechanisms become central components of Transformer blocks, together with positional information and feed-forward processing.

Mark Breakdown

Criterion	Marks
Attention correctly explained	2
Self-attention correctly connected	2
Q/K/V role	2
Scaled dot-product attention	2
Multi-head attention	2
Bridge to Transformers	2
Total	**12**